॥ श्रीगणेशाय नमः ॥

Maxima and Minima

Stationary Point:

A point where the derivative of the considered function is zero.

e.g. $f(x) = x^4 - 4x^3 + 4x^2$

$f'(x) = 4x^3 - 12x^2 + 8x = 0$

$4x(x^2 - 3x + 2) = 0$

$x(x-2)(x-1) = 0$

$x = 0, 1, 2$

stationary points are 0, 1, and 2.

Critical Point:

A point where the derivative of the considered function is either zero or undefined. All stationary points are critical, not the other way around.

e.g. $f(x) = 6x^{4/3} - 3x^{1/3}$

$f'(x) = 8x^{1/3} - x^{-2/3}$

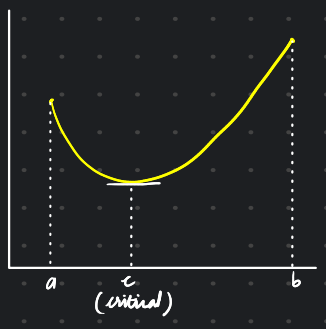
$= 8x^{1/3} - \frac{1}{x^{2/3}} \longrightarrow \frac{8x^{1/3} \cdot x^{2/3} - 1}{x^{2/3}} = 0$

Undefined for $x = 0$. (critical)

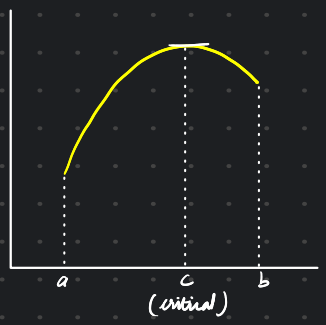
$\Rightarrow 8x - 1 = 0$

$\Rightarrow x = 1/8$ } stationary as well as critical

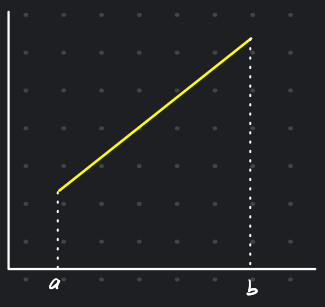
Closed Interval:



- maximum at b
- minima at c



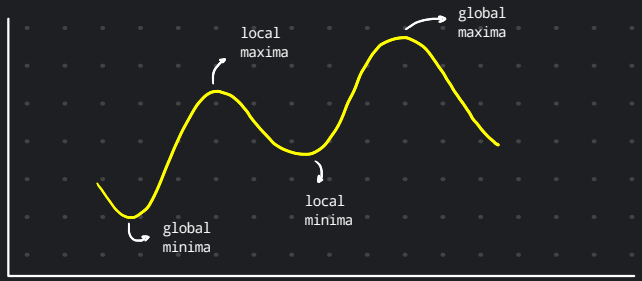
- maximum at c
- minima at a



- maximum at b
- minima at a

Inference:
For closed intervals,
maxima and minima exist for sure (either at the critical or at the end-points).
We don't need the 2nd derivative test.

Extreme value theorem:
If f(x) is continuous in the closed interval [a, b], then f(x) attains an absolute minimum value f(c) and an absolute maximum value f(d) at some c and d in [a, b].



All global maxima are local maxima.
All global minima are local minima.
Not the other way around.

Closed Intervals: Method

- Find the values of f(x) at critical points of f(x) in [a, b].
- Find the values of f(x) at the end points a and b.
- The largest of those is the global maxima, the smallest is the global minima.

Q. $f(x) = x^2 - 4x + 3$ in $[1, 4]$

$f'(x) = 2x - 4 = 0$

$x = 2$

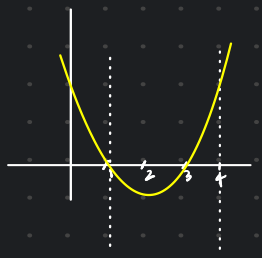
$f(2) = 4 - 8 + 3 = -1$

$f(1) = 0$

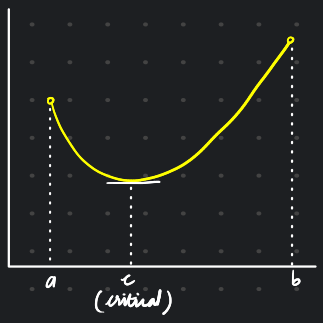
$f(4) = 3$

minima at $x = 2$ (-1)

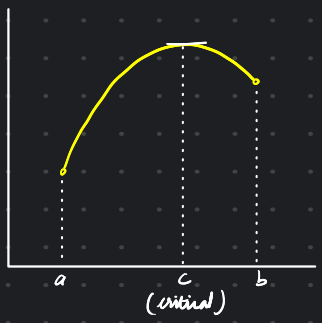
maxima at $x = 4$ (3)



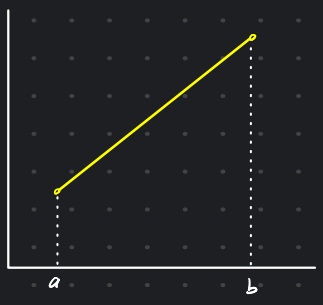
Open Intervals



- maxima doesn't exist
- minima at c

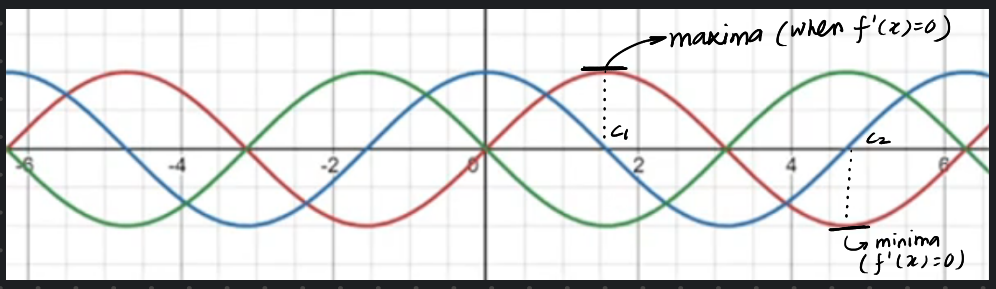


- maxima at c
- minima doesn't exist



- maxima doesn't exist
- minima doesn't exist

In open intervals, if the maxima and minima do exist, both occur at critical points only, never the endpoints.

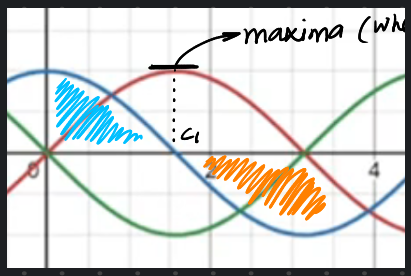


$\cos x = f'(x)$

$\sin x = f(x)$

$-\sin x = -f''(x)$

For x at which f'(x) = 0, how do we say if f(x) is a maxima or a minima?



Here, $f'(x)$ is +ve to the left of maxima.

$f'(x)$ is -ve to the right of maxima.

A point is a maxima when the for the points x to the left of it, f'(x) is +ve, and for those to the right of it, f'(x) is -ve.

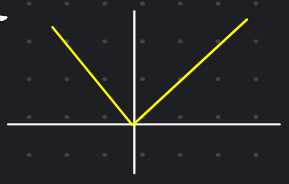
+ve slope → maxima → -ve slope

-ve slope → minima → +ve slope

} 1st derivative test.

Q. $f(x) = |x|$. Find maxima and minima. $(-\infty, \infty)$

→



$f'(x)$ doesn't exist at $x=0$.

0 → critical point. No other critical pts. Can't check at endpoints $(-\infty, \infty)$.

$f(0) = |0| = 0$

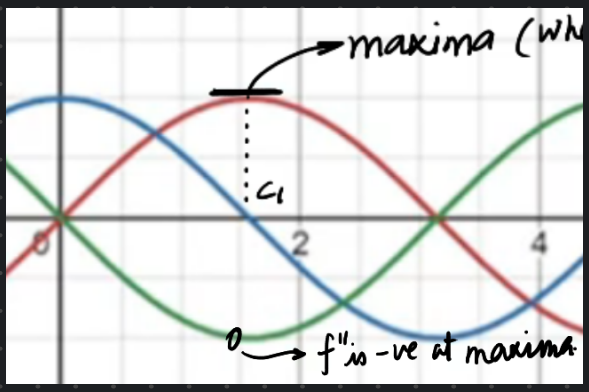
Left → −ve slope (−1)

Right → +ve slope (+1)

} −ve → +ve minima.

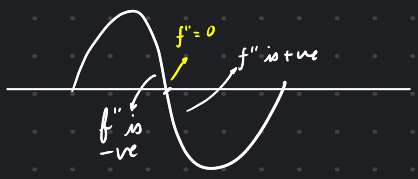
A point is said to be a point of maximum if $f''(c) < 0$ (f'' is −ve → maxima).

A point is said to be a point of minimum if $f''(c) > 0$ (f'' is +ve → minima).

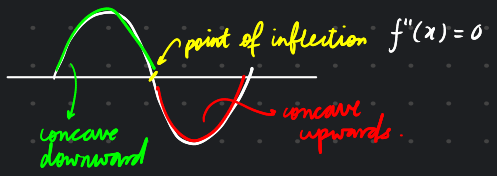


At $f''(x) = 0$, the concavity of $f(x)$ changes.

Points of inflection:
The points where the function's concavity changes.



If the sign of f'' changes from left to right of the point, it is a point of inflection.



Q. $f(x) = x^3$.

→ $f'(x) = 3x^2$

$f'(x) = 0$ at $x=0$. $f(0) = 0$.

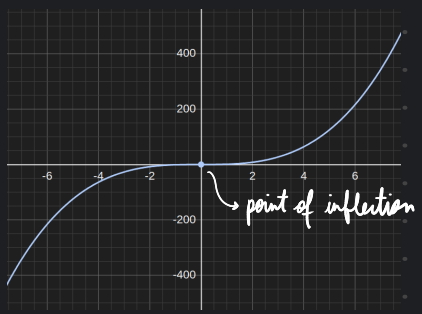
$f''(x) = 6x$

$f''(0) = 0$

$f''(0.01) = 6(0.01) = 0.06$

$f''(-0.01) = 6(-0.01) = -0.06$

f'' changes direction at $x=0$.



Q.

GATE ME 2007

MCQ (Single Correct Answer) +1 -0.3

The minimum value of function $y = x^2$ in the interval $[1, 5]$ is

→

$y = x^2$

$y' = 2x$

$y' = 0$ at $x=0$ not in $[1, 5]$

$f(1) = 1$, $f(5) = 25$ Minima = 1.

Pending stuff (1 1/2 lectures)

GATE 2022 (CE Set-1) Civil Engineering

Q.45	Let max $\{a, b\}$ denote the maximum of two real numbers a and b. Which of the following statement(s) is/are **TRUE** about the function $f(x) = \max\{3 - x, x - 1\}$?
(A)	It is continuous on its domain.
(B)	It has a local minimum at $x = 2$.
(C)	It has a local maximum at $x = 2$.
(D)	It is differentiable on its domain.

✓✓

✓✓

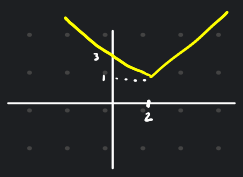
→ $f(x) = \max(3-x, x-1)$

$3-x = x-1$

$x = 2$ → intersecting point

$f(1) = \max(2, 0) = 2$ $\quad f(3) = \max(0, 2) = 2$ $\quad f(4) = (-1, 3) = 3$

$$f(x) = \begin{cases} x-1, & x \geq 2 \\ 3-x, & x < 2 \end{cases}$$



2 → critical pt. ($f'(x)$ not defined)

$f(2) = 1$

slope from -ve → +ve

∴ minima at $x=2$

no maxima

GATE 2022 (CE Set-2) Civil Engineering

Q. 36 - 65 Carry TWO marks each.

Q.36	Consider the polynomial $f(x) = x^3 - 6x^2 + 11x - 6$ on the domain S given by $1 \leq x \leq 3$. The first and second derivatives are $f'(x)$ and $f''(x)$. Consider the following statements: I. The given polynomial is zero at the boundary points $x = \mathbf{1}$ and $x = \mathbf{3}$. II. There exists one local maxima of $f(x)$ within the domain S. III. The second derivative $f''(x) > 0$ throughout the domain S. IV. There exists one local minima of $f(x)$ within the domain S. The correct option is:
(A)	Only statements I, II and III are correct.
(B)	Only statements I, II and IV are correct.
(C)	Only statements I and IV are correct.
(D)	Only statements II and IV are correct.

✓

✓

✗

✓

✓✓

→ $f(x) = x^3 - 6x^2 + 11x - 6$ $\quad S \in [1, 3]$

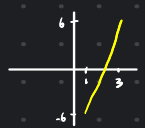
$f'(x) = 3x^2 - 12x + 11$

$f''(x) = 6x - 12$

$f(1) = 1 - 6 + 11 - 6 = 0$

$f(3) = 27 - 54 + 33 - 6 = 0$

I ✓



III ✗

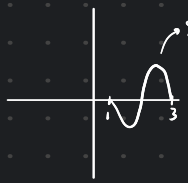
$f'(x) = 0$

$3x^2 - 12x + 11 = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{12 \pm \sqrt{144 - 132}}{6} = \frac{12 \pm \sqrt{12}}{2 \cdot 3} = \frac{6 \pm \sqrt{3}}{3}$

$x = 2 \pm \frac{1}{\sqrt{3}}$

$\approx 1.\infty$ and $2.\infty$



smth. like this

$f''(x) = 6x - 12$

$f''(2 + \frac{1}{\sqrt{3}}) = 12 + \frac{6}{\sqrt{3}} - 12 = \frac{6}{\sqrt{3}} > 0$

∴ $2 + \frac{1}{\sqrt{3}}$ → minima

$f''(2 - \frac{1}{\sqrt{3}}) = 12 - \frac{6}{\sqrt{3}} - 12 = -\frac{6}{\sqrt{3}} < 0$ maxima

Q.26 Which one of the following is NOT a correct statement?

(A) The function $\sqrt[x]{x}$, (x>0), has the global maxima at $x = e$
(B) The function $\sqrt[x]{x}$, (x>0), has the global minima at $x = e$
(C) The function x^3 has neither global minima nor global maxima
(D) The function $|x|$ has the global minima at $x = 0$

GATE 2019 | CE-(Set No 1) Stage 2

→ c) x^3  c is correct d) $|x|$  correct

a) $y = \sqrt[x]{x} = x^{1/x}$

$\ln y = \frac{1}{x}\ln x$

$\frac{1}{y}y' = \frac{1}{x}\frac{1}{x} + \left(\frac{-1}{x^2}\right)(\ln x) = \frac{1-\ln x}{x^2}$

$y' = \frac{x^{1/x}}{x^2}(1-\ln x)$

$(1-\ln x) = 0$

$x = e$

e^{-1} e^2

$f(e^{-1}) = \frac{\frac{1}{e^e}\left(1-\ln\frac{1}{e}\right)}{\frac{1}{e^2}} = \frac{e^2}{e^e}\cdot 2 > 0$

$f'(e^2) = \frac{e^2\sqrt{e^2}\,(1-\ln e^2)}{e^4} = -\frac{e^2\sqrt{e^2}}{e^4} < 0$

+ve → −ve at $x=e$ } $x = e$ is maxima.

a is correct.
b is incorrect.

Q.26 The range of values of k for which the function

$$f(x) = (k^2-4)x^2 + 6x^3 + 8x^4$$

has a local maxima at point $x = 0$ is

(A) $k < -2$ or $k > 2$
(B) $k \le -2$ or $k \ge 2$
(C) $-2 < k < 2$
(D) $-2 \le k \le 2$

GATE 2016 Production and Industrial Engineering (PI)

→ $f(x) = (k^2-4)x^2 + 6x^3 + 8x^4$

$f'(x) = 2(k^2-4)x + 18x^2 + 32x^3$

$f''(x) = 2(k^2-4) + 36x + 96x^2$

Local maxima at $x=0$

$f''(0) < 0$

$2(k^2-4) + 0 + 0 < 0$

$k^2 - 4 < 0$

$k^2 < 4$

$-2 < k < 2$ opt. (C)

If x = -1 and x = 2 are extreme points of f(x) = α log |x| + βx² + x then

(a) α = - 6, β = 1/2

(b) α = - 6, β = -1/2

(c) α = 2, β = -1/2

(d) α = 2, β = 1/2

JEE mains 2018

→ $f(x) = \alpha\log|x| + \beta x^2 + x$

$f'(x) = \alpha\frac{1}{|x|}\frac{|x|}{x} + 2\beta x + 1$

$= \frac{\alpha}{x} + 2\beta x + 1 = 0$

$\frac{\alpha + 2\beta x^2 + x}{x} = 0$

$\alpha + 2\beta x^2 + x = 0$

Critical points ⇒ −1, 2

$\therefore \alpha + 2\beta(-1)^2 + (-1) = 0$ $\alpha + 2\beta(2)^2 + (2) = 0$

$\alpha + 2\beta = 1$ $\alpha + 8\beta = -2$

$\alpha + 2\beta = 1$
$\alpha + 8\beta = -2$
$6\beta = -3$
$\beta = -1/2$

$\alpha + 2\left(-\frac{1}{2}\right) = 1$
$\alpha = 2$

$(\alpha, \beta) \equiv (2, -1/2)$

GATE 2016 — Instrumentation Engineering - IN

Q.26 Let $f:[-1,1]\to\mathbb{R}$, where $f(x)=2x^3-x^4-10$. The minimum value of $f(x)$ is_____. -13

→ $f(x)=2x^3-x^4-10$

$f'(x)=6x^2-4x^3$ $\quad 6x^2-4x^3=0 \Rightarrow x^2(6-4x)=0 \Rightarrow x=0,0,\underbrace{3/2}_{\text{not in } [-1,1]}$

$f''(x)=12x-12x^2$

$f(0)=-10$ min.

$f(-1)=-2-1-10=-13$

$f(1)=2-1-10=-9$ max.

GATE 2016 EE paper Electrical Engineering Set 1 (2nd Stage)

Q.1 The maximum value attained by the function $f(x)=x(x-1)(x-2)$ in the interval [1, 2] is _____. 0

→ $f(x)=x(x^2-3x+2)=x^3-3x^2+2x$

$f'(x)=3x^2-6x+2=0$

$x=\frac{6\pm\sqrt{36-24}}{2\cdot3}=\frac{6\pm\sqrt{4\cdot3}}{2\cdot3}=\frac{3\pm\sqrt{3}}{3}=1\pm\frac{1}{\sqrt{3}}$ $\quad 1-\frac{1}{\sqrt{3}}\notin[1,2]$

$f''(x)=6x-6$

$f(1+\frac{1}{\sqrt{3}})=x(x-1)(x-2)=(1+\text{ve})(1+\text{ve}-1)(1+\text{ve}-2)<0$

$f(1)=0$

$f(2)=0$

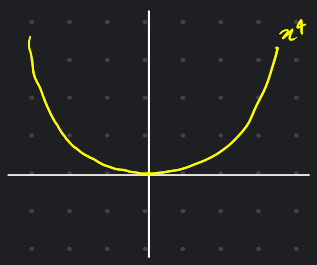
$\max(\text{some -ve \#}, 0, 0)=0$

Q.30 The curve $y=x^4$ is

(A) concave up for all values of x
(B) concave down for all values of x
(C) concave up only for positive values of x
(D) concave up only for negative values of x

GATE 2015 — PRODUCTION & INDUSTRIAL ENGG. – PI

→



$f(x)=x^4$

$f'(x)=4x^3$

$f''(x)=12x^2>0 \ \forall x$

$f''(x)>0 \ \forall x$

$\Rightarrow$ concave up ∪

$f''(x)<0 \ \forall x$

$\Rightarrow$ concave down ∩